

SCR Measure — Surface

Recording slow-rate pump pressures for a surface BOP well

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Well Control

Surface BOP

5 sections

A4 PDF

1. What It Is

This screen records the **Slow Circulating Rate (SCR)** pressures for a Surface BOP well. Before any kill operation, the crew pumps at several slow speeds — typically 20 to 60 strokes per minute — and notes the standpipe pressure at each. Those pressures become reference values when a well kick has to be killed.

It is reached from Operations › Drilling › Well Control › SCR Measurement › Surface, and produces a shareable A4 field sheet as a PDF.

Domain	Oil & gas — well control and drilling
Accent colour	Cyan, matching the SCR tile in the Well Control hub
Output	A4 field sheet PDF through the Share menu
Saving	Three named slots on the device
Also inside	An embedded BOP and mud-pump training simulator

2. The Five Sections

The screen is a stack of five collapsible cards.

#	Section	Contents
1	Well / Job Info	Well name, field, rig, operator, date (pre-filled with today) and engineer
2	Pump Specification	Pump model, liner size, stroke length, strokes-per-minute rating, displacement (automatic), efficiency and maximum pressure
3	SCR Pressure at Speed	A table of up to six readings, each pairing a pump speed with the pressure recorded. Includes the embedded simulator

#	Section	Contents
4	Surface BOP Pressures	Standpipe, casing and choke-line pressures, plus mud weight
5	Summary & Results	The primary and secondary SCR pressures, worked out automatically

3. How to Use It

1. Fill in **Well / Job Info**. The date can be changed.
2. Enter the **pump specification**. Liner size can be typed in inches or millimetres using a unit toggle. Displacement appears by itself.
3. Run the pump at each target speed for **at least three minutes**, then enter the settled speed and standpipe pressure on a row. Up to six rows are available.
4. Enter the **Surface BOP pressures** and the mud weight.
5. Read the **Summary**: it updates as you type.
6. Tap **Save** to store the sheet in one of three named slots, or **Open** to bring one back.
7. Tap **Export PDF** to build the field sheet and share it.

Field procedure note

SCR pressures must be recorded again whenever the mud weight, the pump liner or the drill-string configuration changes. Old SCR figures should never be used in a kill sheet.

4. The Calculations

Result	How it is worked out
Pump displacement (barrels per stroke)	A triplex pump constant, 0.000243, × liner diameter ² × stroke length. If the liner is entered in millimetres, it is first converted to inches
Primary SCR pressure	The lowest recorded pressure, the one at the slowest stable speed
Secondary SCR pressure	The next-lowest recorded pressure, a back-up reading at a slightly higher rate

Both results update as soon as any pressure is typed, and show a dash until at least one reading is entered.

5. The Embedded Simulator

Inside Section 3 is a training widget with three tabs:

- **BOP Control** — shows blowout-preventer valve states as open, blocked or closed.
- **Mud Pumps** — start and stop the pumps and watch the speed and pressure gauges.

- **Driller's Console** — a simulated read-out panel.

6. Saving and the PDF

- **Saving.** Three slots on the device, each showing the name you gave it and the time it was saved.
- **The PDF.** A single A4 page: a header block, a well-identification grid, a pump-specification table, the SCR table with alternating row shading, a notes area, an authorisation and signature block and a page footer.

7. Look and Feel

Colour	Meaning
Pale yellow field	A value you type in
Red field	A value the screen calculates (read-only)
Cyan	Interactive controls for the SCR tools
Green	Collapsible section headings, matching the kill sheets
Orange	The Export PDF button only